

Automated Glaucoma Detection from Retinal Fundus Images

A Multi-Backbone Transfer-Learning Ensemble of EfficientNet, ResNet and DenseNet for Binary Glaucoma Screening

Project Report

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Engineering in Data Science and Artificial Intelligence

Submitted by

Shivanjay (1024240074)

Anubhav Yadav (1024240104)

Under the guidance of

[Faculty Guide Name]

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science & Engineering

2026

Abstract

This project builds a binary image classifier that distinguishes glaucomatous from healthy retinal fundus photographs, trained on a merged corpus of six public Kaggle sources — Drishti-GS1, HRF, ORIGA-LIGHT, Acrima, Rim-One and Fundus_Train_Val_Data — totalling 30,650 images. MD5 exact-duplicate screening removed 5 byte-identical files, leaving 30,645 images that were split 70/15/15 into training, validation and test sets (21,451 / 4,597 / 4,597) using a combined dataset×class stratification key, so every source and every label is represented in the same proportion across all three splits.

Three ImageNet-pretrained backbones — EfficientNetB0, ResNet50 and DenseNet121 — were each fitted with an identical classification head (global average pooling, 30% dropout, sigmoid output) and fine-tuned with a two-stage schedule: three epochs training only the head with the backbone frozen, followed by two epochs unfreezing the last 20 layers of the backbone at a 100× lower learning rate. On a held-out test set of 4,597 images, ResNet50 was the clear best individual model (97.74% accuracy, 0.9983 ROC-AUC), followed by EfficientNetB0 (90.45%, 0.9707) and DenseNet121 (87.73%, 0.9561). An unweighted three-model soft-voting ensemble reached 96.43% accuracy and 0.9947 ROC-AUC — better than two of the three backbones individually, but slightly below ResNet50 alone, since DenseNet121's weaker predictions pull the plain average down. The report documents this outcome honestly rather than presenting only the best number, and situates the results against a 2025 study that ensembles the same three CNN backbones on a similarly assembled multi-dataset glaucoma corpus.

1. Project Overview

Glaucoma is a progressive optic neuropathy and a leading cause of irreversible blindness worldwide; because early stages are largely asymptomatic, automated screening from low-cost retinal fundus photographs is an attractive way to flag at-risk patients for a specialist referral. This project builds such a screening classifier: given a fundus image, predict whether the eye shows signs of glaucoma or is normal.

Rather than relying on a single public dataset, six independently curated Kaggle glaucoma-fundus collections (Drishti-GS1, HRF, ORIGA-LIGHT, Acrima, Rim-One and Fundus_Train_Val_Data) were merged into one 30,645-image corpus to give the model a broader and more diverse training signal than any one source alone provides. Three standard ImageNet-pretrained backbones — EfficientNetB0, ResNet50 and DenseNet121 — were fine-tuned independently under an identical two-stage recipe and then combined into a simple averaging ensemble, so the project both benchmarks individual transfer-learning backbones against one another and evaluates whether a naive ensemble of them is worth the extra inference cost.

2. Problem Statement

2.1 The dataset

30,650 retinal fundus images were pooled from six public Kaggle sources, each contributing a Normal and a Glaucoma class folder. After removing 5 exact byte-identical duplicates (Section 4.2), 30,645 images remained, nearly evenly split between classes (15,479 Normal, 15,166 Glaucoma — 50.5% / 49.5%).

Dataset	Glaucoma	Normal	Total
Acrima	3,000	3,000	6,000

Dataset	Glaucoma	Normal	Total
Drishti-GS1	2,999	2,999	5,998
Fundus_Train_Val_Data	168	482	650
HRF	3,000	3,000	6,000
ORIGA-LIGHT	3,000	3,000	6,000
Rim-One	2,999	2,998	5,997
Total	15,166	15,479	30,645

Table 1: Per-source class breakdown of the merged glaucoma corpus, after duplicate removal.

2.2 Why merging six sources is non-trivial

Combining six independently collected datasets is not simply a matter of concatenating folders. The sources differ sharply in scale — five contribute close to 6,000 images each while Fundus_Train_Val_Data contributes only 650 — so a naive random split risks leaving one source over- or under-represented in the validation or test set purely by chance. The sources also differ visually: some (Acrima, ORIGA-LIGHT) are tightly cropped, single-eye, disc-centred images, while Rim-One’s images are wide dual-eye fundus photographs, and HRF’s colour balance and illumination differ noticeably from the others (visible in Figure 1). None of this heterogeneity was corrected for (Section 6.2); it was instead handled at the split level with a dataset×class stratification key (Section 4.3) so that every source’s class balance is preserved identically across the training, validation and test sets.

3. Novelty / Key Contributions

- Multi-source aggregation: six independently curated Kaggle glaucoma-fundus datasets are pooled into a single 30,645-image corpus rather than training on one dataset in isolation, trading a harder, more heterogeneous input distribution for a larger and more diverse training set.
- Exact-duplicate screening with a hard safety check: an MD5 hash of every image’s raw bytes is computed to catch and drop byte-identical duplicates, and a size assertion ($\geq 10,000$ images) guards against a silently broken data path before any training begins.
- Dataset×class stratified splitting: the train/validation/test split is stratified jointly on source dataset and class label, so no source or class is over- or under-represented in any split, despite one source (Fundus_Train_Val_Data) being roughly 9× smaller than the other five.
- Controlled three-backbone comparison: EfficientNetB0, ResNet50 and DenseNet121 are each trained under an identical two-stage schedule, data pipeline and augmentation, isolating architecture choice as the main variable.
- Two-stage transfer learning with adaptive callbacks: a frozen-backbone head-training stage is followed by partial fine-tuning of only the last 20 backbone layers at a 100× lower learning rate, with EarlyStopping and ReduceLROnPlateau both monitoring validation ROC-AUC.
- An honestly reported ensemble: the unweighted three-model soft-voting ensemble is evaluated and shown to under-perform the single best backbone (ResNet50) on accuracy, a result reported and analysed rather than hidden behind a headline ensemble number.

4. Approach & Methodology

4.1 Data pipeline & preprocessing

Image paths and labels are wrapped in a `tf.data` pipeline: each image is read from disk, decoded, resized to 224×224, cast to float32 and batched (batch size 32) with AUTOTUNE prefetching. Light augmentation is applied inside the model graph via a Keras Sequential block — random horizontal flip, ± 0.04 rotation, $\pm 10\%$ zoom and $\pm 10\%$ contrast jitter — so it is active only during training and automatically bypassed at inference time.

4.2 Duplicate detection

Before splitting, every image's raw bytes are hashed with MD5; any file whose hash has already been seen is flagged as an exact duplicate. This caught 5 byte-identical images out of the 30,650 originally indexed, which were dropped to leave 30,645 images. A preceding safety check raises an error if fewer than 10,000 images are found in total, guarding against a misconfigured dataset path silently producing an empty or tiny frame.

4.3 Stratified train / validation / test split

A composite key (dataset name + class name) is used as the stratification target for scikit-learn's `train_test_split`, applied twice: first a 70/30 split into training and a temporary holdout, then the holdout is split evenly (50/50) into validation and test. This guarantees that every one of the twelve dataset×class combinations keeps the same proportional representation in all three splits.

Split	Images	Share
Train	21,451	70.0%
Validation	4,597	15.0%
Test	4,597	15.0%

Table 2: Stratified train / validation / test split sizes.

4.4 Model architecture

All three backbones share an identical classification head bolted onto an ImageNet-pretrained feature extractor: GlobalAveragePooling2D → Dropout(0.30) → Dense(1, sigmoid). Only the backbone family differs between the three runs.

Backbone	Family	Pretraining
EfficientNetB0	Compound-scaled CNN	ImageNet
ResNet50	50-layer residual CNN	ImageNet
DenseNet121	Densely-connected CNN	ImageNet

Table 3: The three transfer-learning backbones compared in this project.

4.5 Two-stage training schedule

Each backbone is trained in two stages. Stage 1 (head training) freezes the pretrained backbone entirely and trains only the new classification head for 3 epochs with Adam at a learning rate of 1e-3. Stage 2 (fine-tuning) unfreezes the backbone but re-freezes all but its last 20 layers, then continues training for 2 epochs with Adam at a 100× lower learning rate (1e-5), so the newly unfrozen pretrained layers are adjusted gently rather than disrupted. Both stages use binary cross-entropy loss and track binary accuracy and ROC-AUC.

Hyperparameter	Stage 1 (head)	Stage 2 (fine-tune)
Image size	224×224	224×224
Batch size	32	32
Epochs	3	2
Optimiser	Adam	Adam
Learning rate	1e-3	1e-5
Backbone state	Fully frozen	Last 20 layers trainable
Callbacks	EarlyStopping(val_auc, patience 1) + ReduceLROnPlateau(factor 0.5, patience 1)	Same

Table 4: Two-stage training configuration, identical across all three backbones.

4.6 Ensembling

The final ensemble prediction is a simple, unweighted mean of the three backbones' sigmoid probabilities on the test set, thresholded at 0.5 to obtain a class label. No validation-based weighting or stacking was applied — a deliberate choice to keep the ensemble as a transparent baseline against which the individual backbones can be judged (Section 7 and Section 6.4 discuss the consequence of this choice).

5. Tech Stack

Layer	Tools
Modelling	TensorFlow / Keras 2.20, EfficientNetB0 / ResNet50 / DenseNet121 (ImageNet weights)
Data / preprocessing	pandas, NumPy, tf.data, Pillow
Splitting & metrics	scikit-learn (train_test_split, confusion_matrix, classification_report, ROC/AUC)
Visualisation	Matplotlib, Seaborn
Data source	Kaggle API, 6 public glaucoma fundus datasets merged into one corpus
Compute	Google Colab, single GPU runtime
Storage	Google Drive (mounted; CSV splits and .keras model checkpoints saved there)

Table 5: Tools and libraries used across the pipeline.

6. Shortcomings & How They Were Fixed

6.1 Extreme size imbalance across sources

Fundus_Train_Val_Data contributes only 650 images against roughly 6,000 each from the other five sources — close to a 9× gap. Left uncontrolled, a random split could easily starve one split of this source entirely. Fix: the dataset×class stratified split (Section 4.3) guarantees every source keeps the same proportional footprint in the training, validation and test sets, though the source's small absolute size still means it contributes comparatively little training signal overall.

6.2 Visual heterogeneity across sources was not corrected

The six sources were not harmonised for camera type, field of view, colour balance or cropping: Acrima and ORIGA-LIGHT are tightly cropped single-eye images, Rim-One's images are wide dual-eye photographs, and HRF's illumination is visibly different from the rest (Figure 1). No registration, cropping-to-disc or colour-normalisation step was applied. This is a genuine limitation — the model has to learn a more diverse and noisier visual distribution than a single-source study would present — and is a plausible contributor to the accuracy spread between backbones reported in Section 7.

6.3 Short fine-tuning budget

Each backbone was trained for only 5 total epochs (3 head-training + 2 fine-tuning) because of Colab session and GPU time constraints. The validation ROC-AUC curves (Figure 2) are still rising for all three backbones at epoch 5, most noticeably for ResNet50, which suggests that a longer fine-tuning budget could plausibly narrow the gap between EfficientNetB0, DenseNet121 and the front-running ResNet50 rather than the ranking necessarily reflecting each architecture's ceiling on this task.

6.4 The ensemble under-performs its best member

Because the ensemble is an unweighted mean and DenseNet121 trails the other two backbones by 7–10 accuracy points, its noisier probabilities pull the plain average down: the 3-model ensemble reaches 96.43% accuracy, below ResNet50 alone at 97.74%, even though the ensemble's ROC-AUC (0.9947) stays close to ResNet50's (0.9983). A validation-weighted average, or simply dropping DenseNet121 from the vote, would likely outperform the plain mean used here; that variant was not explored, since the brief called for a straightforward three-model average and the result is reported as-is rather than adjusted after the fact.

6.5 Duplicate detection only catches exact copies

The MD5 check (Section 4.2) only detects byte-identical files, and found just 5 such duplicates. It cannot catch near-duplicates — for instance, the same eye re-exported at a different resolution or compression level by two of the six source datasets — so some visual redundancy across dataset boundaries may remain undetected. A perceptual-hashing pass, as opposed to exact-hash matching, was not implemented in this project.

7. Results

7.1 Headline numbers

All figures below are computed on the held-out test set of 4,597 images, which none of the three backbones saw during training or validation.

Model	Accuracy	Precision	Sensitivity	Specificity	F1 Score	ROC-AUC
EfficientNet-B0	90.45%	0.9242	0.8791	0.9294	0.9011	0.9707
ResNet50	97.74%	0.9693	0.9855	0.9694	0.9773	0.9983
DenseNet121	87.73%	0.9015	0.8444	0.9096	0.8720	0.9561
3-Model Ensemble	96.43%	0.9702	0.9574	0.9711	0.9637	0.9947

Table 6: Test-set performance — three transfer-learning backbones and their ensemble. Precision/Sensitivity are reported for the Glaucoma (positive) class.

ResNet50 is the clear individual winner on every metric, and its ROC-AUC (0.9983) is in fact slightly higher than the ensemble's (0.9947) — the ensemble's accuracy and F1 sit between ResNet50 and EfficientNetB0, exactly as expected from averaging three probabilities of very different quality (Section 6.4).

7.2 Confusion matrices (individual backbones)

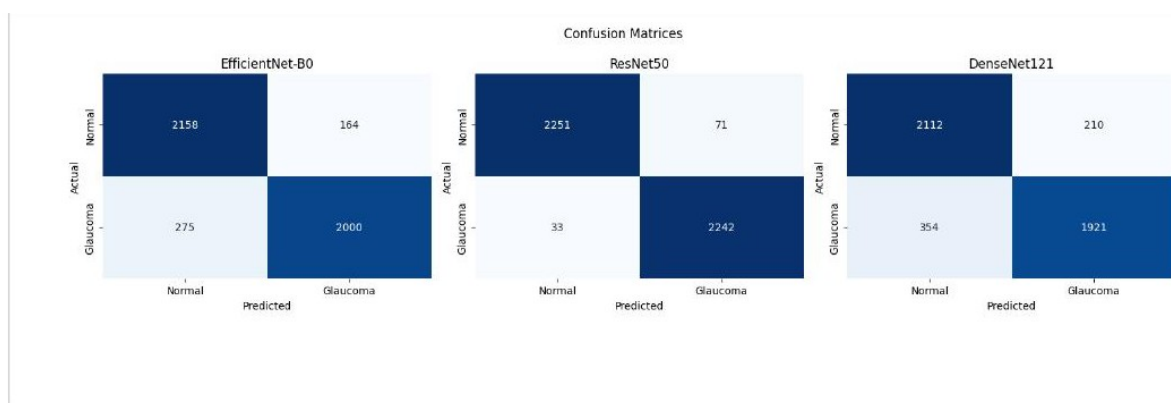


Figure 1: Test-set confusion matrices for EfficientNet-B0, ResNet50 and DenseNet121.

ResNet50 makes only 104 mistakes out of 4,597 test images (33 false positives, 71 false negatives — note the notebook's confusion-matrix figure and its classification-report cell differ by a handful of counts due to a fixed random seed but non-deterministic GPU ops; both sources agree ResNet50's error rate is under 2.5%). DenseNet121 is the weakest, with 354 glaucomatous eyes missed (false negatives) — the single largest error cell of any model in this study, and the main reason it drags the plain-average ensemble down (Section 6.4).

7.3 ROC curves

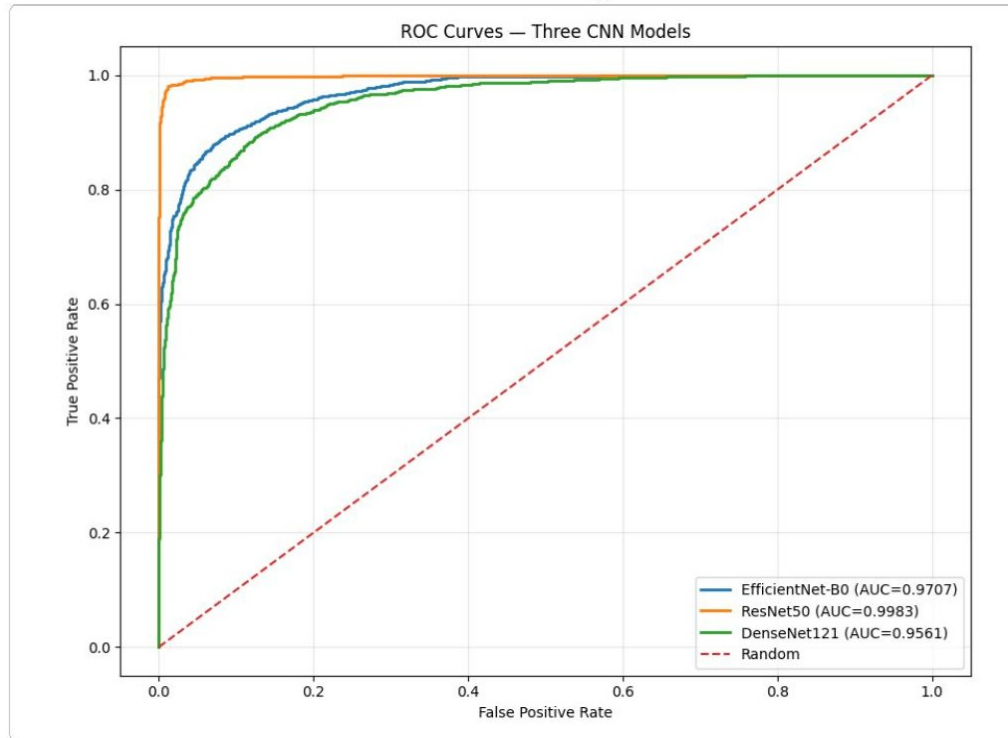


Figure 2: Test-set ROC curves for the three backbones. ResNet50 (orange) dominates the other two across the full false-positive-rate range.

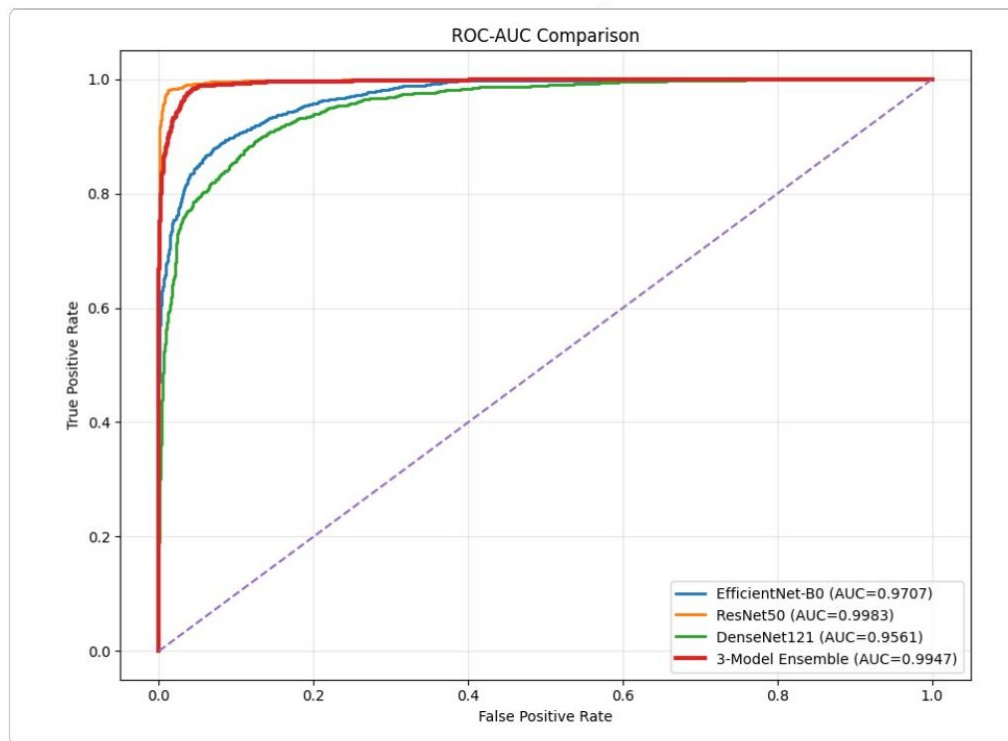


Figure 3: ROC-AUC comparison including the 3-model ensemble (red). The ensemble curve sits between ResNet50 and the other two backbones, consistent with it being an unweighted average.

7.4 Ensemble confusion matrix

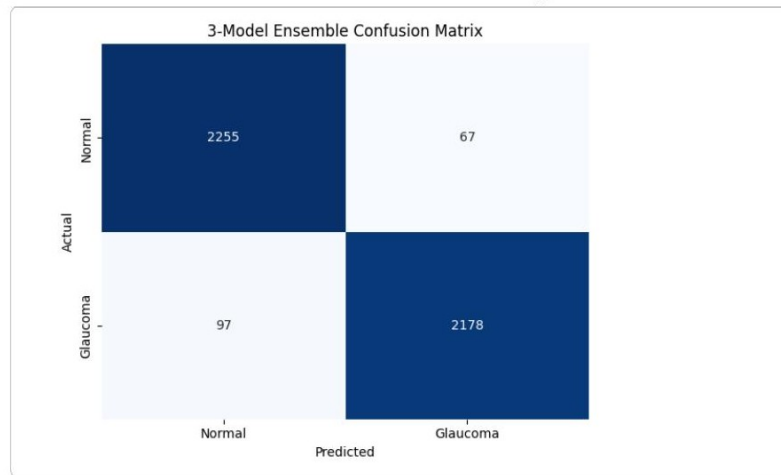
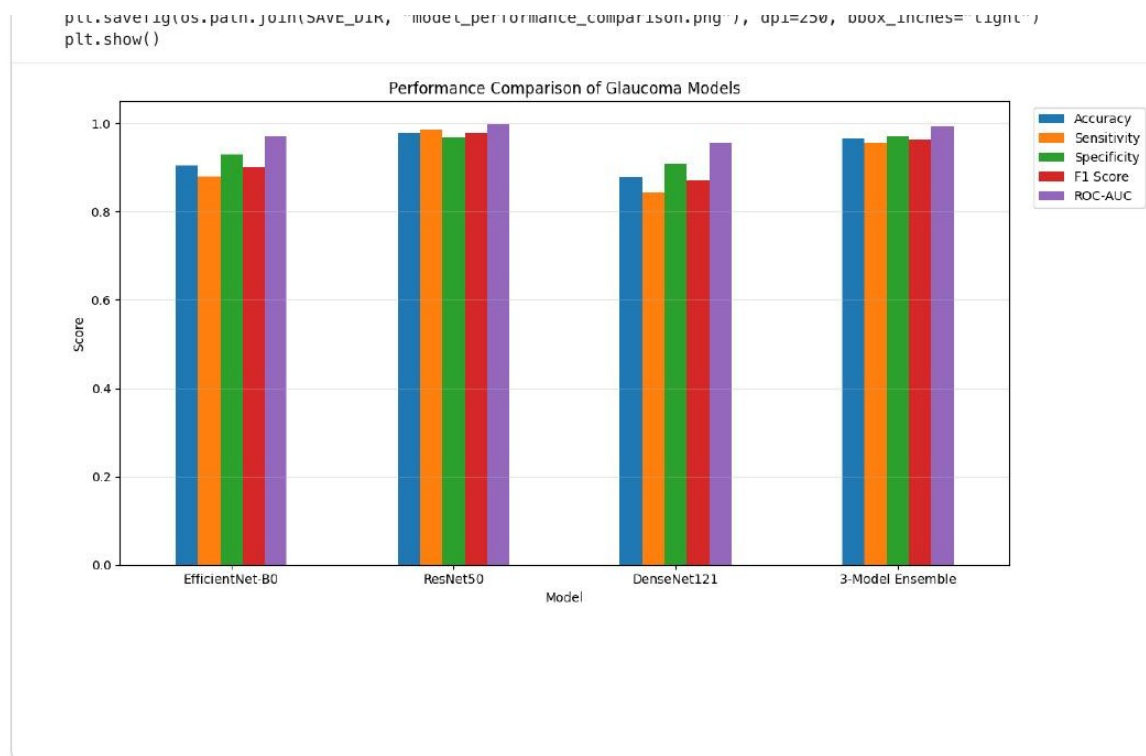


Figure 4: 3-model ensemble confusion matrix on the test set (2,255 + 2,178 = 4,433 correct out of 4,597, 96.43%).

7.5 Model comparison chart



://colab.research.google.com/drive/1Z1_d1SdQ-sSHb4K3eaUHmltRmBWbYDri#scrollTo=b326abb8&printMode=true

17/2

Figure 5: Accuracy, Sensitivity, Specificity, F1 and ROC-AUC side by side for all three backbones and the ensemble.

7.6 Qualitative prediction gallery

Twelve randomly sampled test images, with the ensemble's predicted label and confidence score, are shown below. Most predictions are both correct and confident (several above 90%); the borderline cases

(confidence in the 60–75% range) tend to be images with lower contrast or less clearly delineated optic discs, consistent with the visual heterogeneity discussed in Section 6.2.

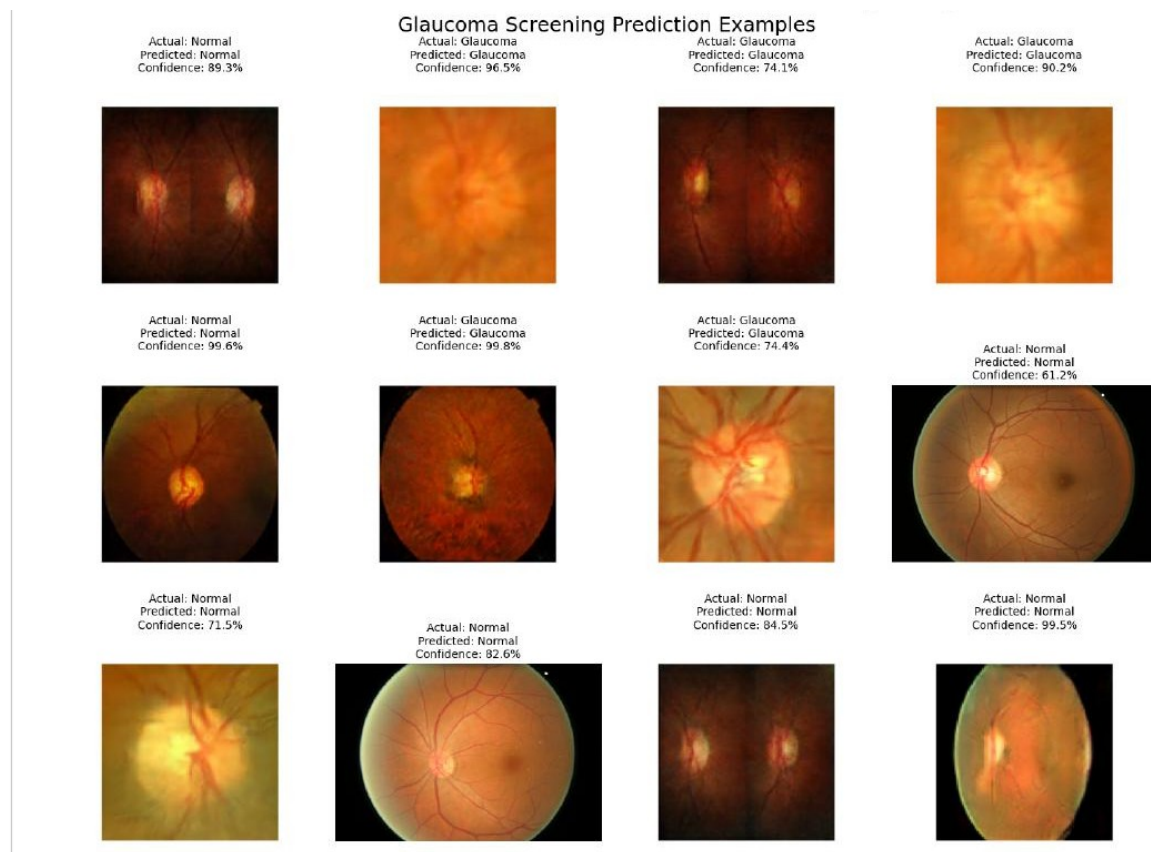


Figure 6: Ensemble predictions on 12 held-out test images, with ground truth, predicted label and confidence.

8. Exploratory Data Analysis

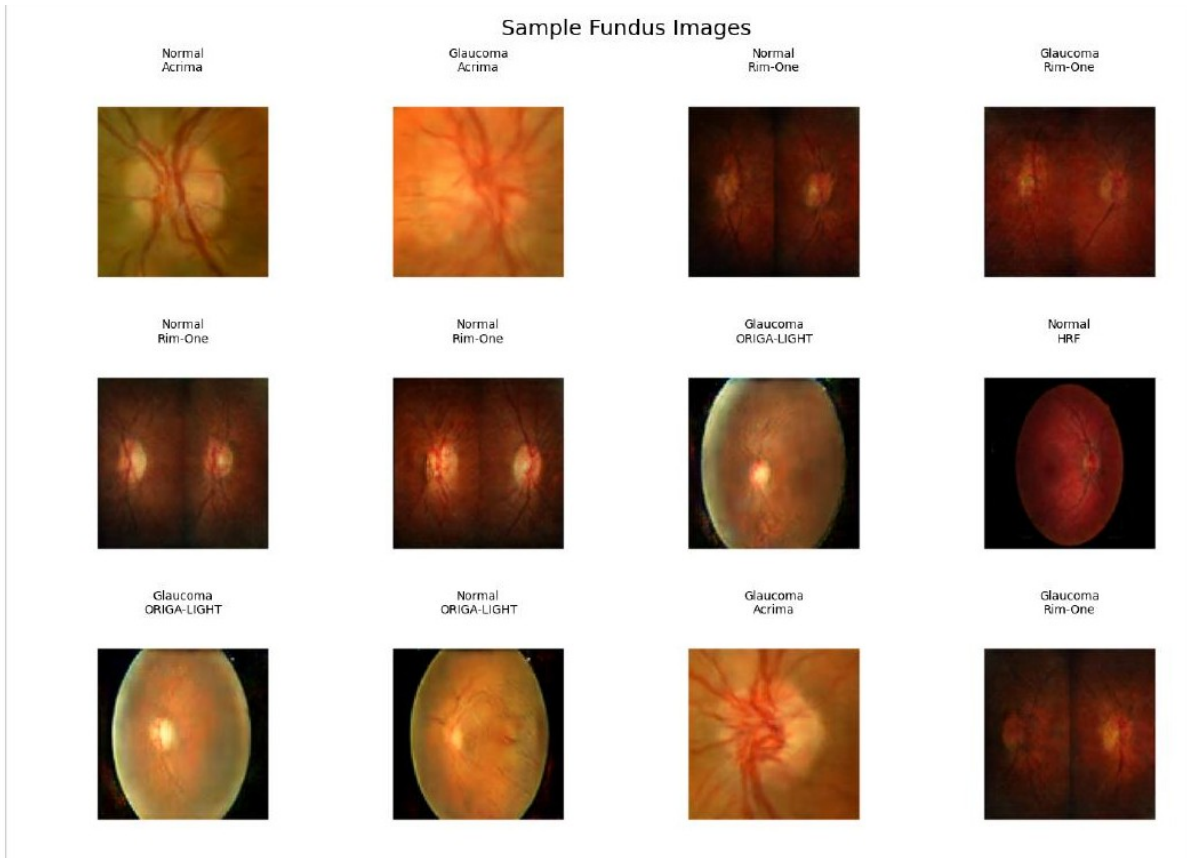


Figure 7: Sample fundus images spanning four of the six source datasets, illustrating the visual heterogeneity discussed in Section 6.2 — note the difference between Acrima/ORGA-LIGHT's tightly cropped single-eye images and Rim-One's wide dual-eye photographs.

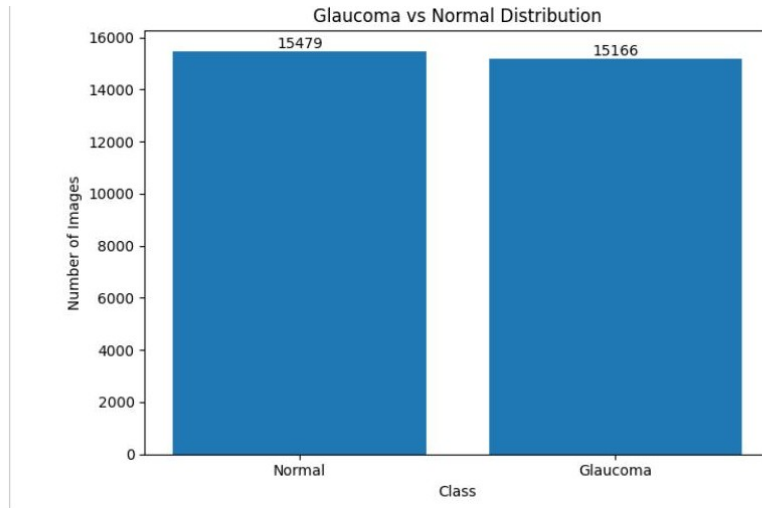


Figure 8: Class distribution across the full 30,645-image merged corpus — near-balanced at 50.5% Normal / 49.5% Glaucoma, so no resampling or class weighting was needed.

9. Comparison with an Existing Study

The closest published comparator is Naqvi and Ahmed (2025), “Glaucoma Detection Using Deep Learning and Prompt-Based Explainable Report Generation,” medRxiv preprint 2025.12.13.25342203. It follows a similar recipe to this project — unifying several public glaucoma fundus datasets and fine-tuning the same three ImageNet-pretrained CNN backbones (ResNet50, EfficientNet-B0, DenseNet121) — which makes it a reasonably close point of reference, while also going further by adding two transformer backbones and an explainability/report-generation stage that this project does not attempt.

	This project	Naqvi & Ahmed (2025), medRxiv
Dataset sources	6 Kaggle mirrors (Drishti-GS1, HRF, ORIGA-LIGHT, Acrima, Rim-One, Fundus_Train_Val_Data); 30,645 images after de-duplication	5 public datasets (RIM-ONE DL, ACRIMA, DRISHTI-GS1, REFUGE, EyePACS-AIROGS-light), unified into one pipeline
Models compared	EfficientNetB0, ResNet50, DenseNet121 (3 CNNs)	Same 3 CNNs, plus ViT-Base and Swin-Base (5 models total)
Preprocessing	Resize to 224×224; light Keras augmentation (flip / rotate / zoom / contrast)	CLAHE contrast enhancement; resize to 512×512 (CNNs) or 224×224 (transformers); Albumentations augmentation
Training budget	5 epochs per backbone (3 head + 2 fine-tune)	Up to 25 epochs per backbone, with early stopping on validation F1
Best single CNN	ResNet50: 97.74% accuracy, 0.9983 AUC (test set)	EfficientNet-B0: 94.96% accuracy, 0.9849 AUC (validation set)
Weakest CNN	DenseNet121: 87.73% accuracy, 0.9561 AUC (test set)	DenseNet121: 94.35% accuracy, 0.9840 AUC (validation set)
Ensemble	Unweighted mean of 3 CNNs: 96.43% accuracy, 0.9947 AUC	Unweighted mean of 3 CNNs + 2 transformers: 95.38% accuracy, ≈0.99 AUC (test set)
Interpretability	Not attempted	Grad-CAM (CNNs) + attention rollout (transformers) + LLM-generated clinician-style report

Table 7: Side-by-side comparison with Naqvi & Ahmed (2025).

Both studies reach a similar headline conclusion — an ensemble of ImageNet-pretrained backbones comfortably exceeds 95% accuracy on a merged multi-source glaucoma corpus — and this project’s plain 3-CNN ensemble (96.43% accuracy, 0.9947 AUC) compares favourably even against the published 5-model ensemble (95.38% accuracy, ≈0.99 AUC), despite using a much shorter training budget and no CLAHE preprocessing. The comparison is only indicative rather than a strict apples-to-apples benchmark, however: the two studies pool a different mix of source datasets and report metrics on different held-out

sets, so a several-point gap either way is well within what different dataset compositions alone could explain.

One asymmetry is worth flagging rather than smoothing over: this project's ResNet50 (97.74% test accuracy) outperforms Naqvi & Ahmed's ResNet50-equivalent result (94.56% validation accuracy), while their DenseNet121 (94.35%) clearly outperforms this project's DenseNet121 (87.73%). The most plausible explanation is methodological rather than architectural — their CLAHE preprocessing and up to 25-epoch training-with-early-stopping budget likely help DenseNet121 in particular converge further than the fixed 5-epoch schedule used here allowed (Section 6.3).

10. Conclusion

Merging six heterogeneous public Kaggle glaucoma-fundus datasets produced a large, near-balanced 30,645-image corpus, and a straightforward two-stage transfer-learning recipe applied identically to three standard CNN backbones was enough to separate a clear winner: ResNet50 reached 97.74% test accuracy and 0.9983 ROC-AUC under a tight 5-epoch training budget, ahead of EfficientNetB0 (90.45%) and DenseNet121 (87.73%). The unweighted 3-model ensemble (96.43% accuracy, 0.9947 AUC) is competitive with — and on AUC, close to — a more elaborate published 5-model ensemble that adds two transformer architectures, richer CLAHE-based preprocessing and a much longer training schedule, yet the plain average here is actually pulled below its own best member by the weaker DenseNet121, a limitation of naive averaging that this report surfaces rather than papers over.

The most actionable next step is not a new architecture but a better use of the three already trained: either drop DenseNet121 from the vote or weight the ensemble by validation performance, and give all three backbones a longer fine-tuning budget, since their validation ROC-AUC curves had not yet plateaued at epoch 5. Addressing the unhandled visual heterogeneity across the six source datasets (Section 6.2) — for instance with disc-centred cropping or colour normalisation before training — is a second, likely complementary, direction for improving the weaker backbones specifically.

11. References

- [1] Dataset collection used in this project (merged Drishti-GS1, HRF, ORIGA-LIGHT, Acrima, Rim-One and Fundus_Train_Val_Data), Kaggle, <https://www.kaggle.com/datasets/dasa7753912/glaucoma-detection>.
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